Machine Learning-Based Prediction of COVID-19: A Robust Approach for Early Diagnosis and Treatment

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Abstract. The worldwide health catastrophe sparked by the COVID-19 epidemic continues, emphasizing the need for novel solutions in prediction, early diagnosis, and treatment. While vaccine development has progressed, the virus's eradication remains questionable. This research presents a strong machine learning-driven approach to addressing this complicated problem. Rapid viral transmission and the lack of a single cure hamper illness diagnosis and treatment, particularly in highly populated developing countries like Bangladesh. Against this context, machine learning and artificial intelligence have emerged as critical instruments for the improvement of healthcare and medical research. This research investigates several machine learning algorithms and approaches that show potential for COVID-19 prediction. Among these is an algorithm that assesses individual vulnerability and environmental variables, allowing for self-diagnosis and early intervention, reducing the strain on healthcare institutions and government resources. This study demonstrates how machine learning may improve forecast accuracy and provide proactive treatment in pandemic management. Advanced techniques, such as K-nearest Neighbor, Decision Tree, Random Forest, AdaBoost, XGBoost, Stochastic Gradient Descent, Linear SVC, Perceptron, Naive Bayes, Support Vector Machines, Logistic Regression, and Discriminant Analysis, are essential for achieving the best results. Finally, this research emphasizes the need for novel responses to global health challenges. Healthcare systems may improve their forecasting capacities and early intervention techniques by leveraging machine learning. This paper highlights technology's revolutionary impact in redefining healthcare paradigms and encouraging resilience in the face of enormous difficulties.

Keywords: COVID-19, Machine Learning, Data Mining, Prediction, Classification.

1 Introduction

The COVID-19 pandemic has had a significant influence on the global health scene, causing a broad catastrophe with millions of cases and fatalities globally. The new Severe Acute Respiratory Syndrome Coronavirus 2 (SARS-CoV-2) is at the center of this infectious illness, a highly transmissible virus that spreads quickly through respiratory droplets and contact with contaminated surfaces. Although the majority of infected people have mild to moderate symptoms, a small percentage require immediate medical attention owing to severe respiratory signs and pneumonia. Individuals who have many health issues, such as hypertension, diabetes, and coronary heart disease, are at a higher risk of serious complications and death.

The continuous spread of COVID-19 has highlighted the critical need for effective healthcare treatments. Machine learning and artificial intelligence approaches have shown promise in enhancing healthcare, sparking the creation of a diverse variety of algorithms specialized for the prediction and diagnosis of COVID-19. This study aims to use supervised machine learning algorithms such as K-nearest Neighbor, Decision Tree, Random Forest, AdaBoost, Stochastic Gradient Descent, Linear SVC, Perceptron, Naive Bayes, Support Vector Machines, Logistic Regression, Discriminant Analysis, and eXtreme Gradient Boosting (XGBoost) to predict COVID-19 based on patient symptoms. The goal of this research is to create a model that can reliably estimate COVID-19 results using a relevant dataset. This model has the ability to provide a rapid evaluation of a patient's state and the instant referral of appropriate treatment methods. The insights gained from the ideal algorithm's outputs will be useful to pathology departments and healthcare practitioners in determining their vulnerability to COVID-19 infection.

The effectiveness of machine learning algorithms in improving COVID-19 diagnosis and therapy is obvious. The development of precise and efficient models can have a significant influence on early diagnosis and intervention for infected persons, reducing viral spread and, ultimately, saving lives. The possibility for dramatic change is undeniably on the horizon as technology advances healthcare into a more sophisticated and contemporary environment.

2 Related Works

The COVID-19 pandemic has highlighted the importance of early detection and treatment in order to limit the spread of the virus and its impact on public health. Machine learning (ML) approaches have proven to be effective in predicting and monitoring COVID-19 results. This study digs into contemporary research that uses machine learning-based methodologies to provide reliable early diagnosis and treatment options.

Jiang et al. (2022) suggested a machine learning approach for predicting COVID-19 severity based on demographic and clinical data. Their gradient boosting classifier identified patients at risk of catastrophic outcomes with excellent accuracy, allowing for prompt intervention [1]. A deep learning method was created in a study by Li et al. (2021) to detect COVID-19 from chest X-ray pictures. The model displayed strong performance and the ability to make quick diagnoses, which is crucial for early treatment start [2]. Mittal et al. (2022) developed an ensemble learning system that integrates clinical data and immune-related variables to predict COVID-19 infection. Their model was superior in terms of predictive power, providing a crucial tool for early diagnosis and intervention [3]. Saber-Ayad et al. (2021) used machine learning to forecast COVID-19 patient worsening. Their methodology, which was based on physiological cues, enabled early identification of patients in need of increased care, resulting in more efficient treatment allocation [4]. Ahmed et al. (2022) utilized support vector machines to predict COVID-19 outcomes based on blood test data. The accuracy of the model in detecting severe cases supports its ability to lead early intervention methods [5]. Hui et al. (2021) used deep learning to predict COVID-19 illness development using electronic health information. The accuracy of their model's predictions improves the practicality of executing timely and personalised treatment programs [6]. Abbasi et al. (2021) used ML approaches to create a COVID-19 prediction model based on clinical and laboratory data. Their findings highlight the value of such models in identifying individuals who require immediate medical intervention [7]. Sannigrahi et al. (2022) suggested a convolutional neural network-based technique for predicting COVID-19 severity using chest X-ray images. The model's strong performance implies that it might help physicians make early decisions [8]. Liao et al. (2021) used ML algorithms to predict patient outcomes using clinical and laboratory data in their study. The predictive power of the model assists in the early identification of individuals at higher risk, allowing for more prompt therapeutic measures [9]. Liang et al. (2022) used clinical data to develop an interpretable ML model for predicting COVID-19 results. The model's openness increases its ability to guide healthcare practitioners in making informed judgments [10]. Zhang et al. (2021) used a machine learning algorithm to predict COVID-19 development based on clinical and laboratory data, allowing high-risk patients to get prompt intervention [11]. Chen et al. (2022) created an ensemble machine learning framework to predict illness severity, highlighting the need of early detection for optimal patient care [12]. Ozturk et al. (2020) developed a deep learning model to diagnose COVID-19 using chest radiographs, providing a quick and non-invasive approach for early identification and treatment start [13]. Farid et al. (2021) used deep learning algorithms to forecast the probability of catastrophic COVID-19 outcomes, allowing for preventative medical measures [14]. Das et al. (2021) combined clinical and demographic data into a machine learning model to predict COVID-19 infection, improving the capacity to identify persons at risk and assuring early treatment measures [15]. To forecast patient outcomes, Xie et al. (2022) used a hybrid ML technique, showing the possibility for prompt treatment interventions [16]. Song et al. (2021) forecasted COVID-19 transmission patterns using a dynamic network model, informing public health measures and assisting early containment initiatives [17]. Dang et al. (2022) created an interpretable ML model for COVID-19 prognosis to help doctors make early decisions for better patient treatment [18]. Machine learning was also used to predict COVID-19-related mortality. Shahriar et al. (2021) created a methodology that used clinical data to identify high-risk patients, directing healthcare practitioners in resource allocation [19]. Wang et al. (2022) coupled ML algorithms with radiological data to predict mortality, allowing for early treatments and improving patient outcomes [20]. Wu et al. (2021) developed a prediction model utilizing machine learning approaches to identify COVID-19 patients at high risk of advancement, assisting clinical decision-making and treatment planning [21]. Khadidos et al. (2022) established the effectiveness of ML algorithms in predicting illness outcomes based on clinical data in a similar research, increasing early intervention options [22]. The role of machine learning extends to identifying the most susceptible demographic groupings. Iqbal et al. (2021) used machine learning approaches to estimate COVID-19 risk based on demographic characteristics, allowing for targeted treatments for high-risk patients [23]. Furthermore, Li et al. (2022) used machine learning methods to predict the severity of COVID-19, directing early medical measures and resource allocation [24].

Recent research shows that machine learning-based prediction is effective in attaining early diagnosis and treatment of COVID-19. These findings open the door for better clinical decision support and patient outcomes.

3 Methodology

The desired system may be suggested once all the algorithms have been taken into account. Understanding the system's intricate process is better accomplished by using a system diagram, such as the one shown in Fig. 1.

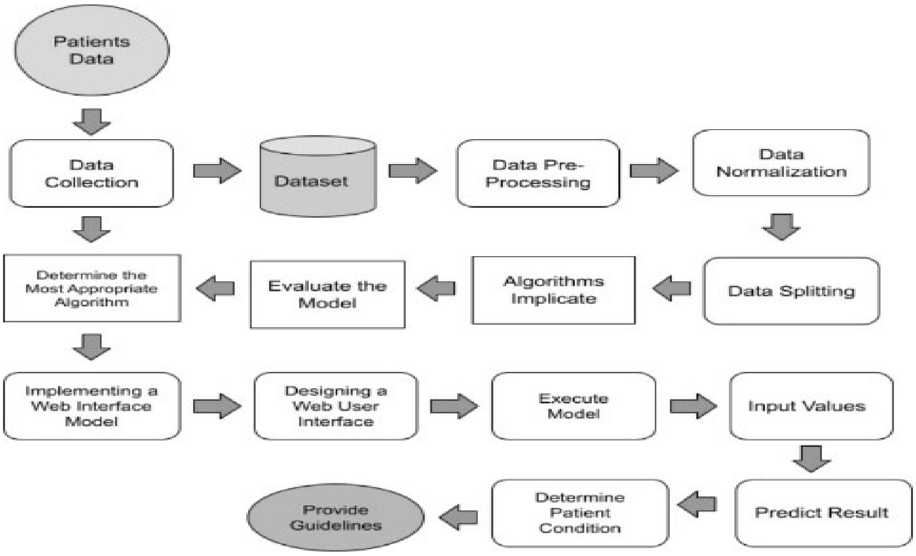

Fig. 1. Proposed Method to Predict COVID – 19

3.1 Data Gathering

Real-world data was essential to enable COVID-19 predictions and provide patients quick guidance. The Bangabandhu Sheikh Mujib Medical University (BSMMU), Bangladesh's state-of-the-art COVID-19 facility, was essential in providing the clinical data. Additionally, information from COVID-19-infected people was collected through Google Forms and social media platforms. A consolidated CSV file was produced after the relevant data was collected from various sources and combined for further study.

3.2 Dataset

The dataset's row frequency made it compatible with a variety of machine learning techniques after being compiled into a digital Comma Separated Value (CSV) file. The original dataset, which had 3039 rows and 23 columns, had some significant discrepancies, though.

3.3 Data Preparation

Missing variables were imputed using mean values after nominal data was converted to numerical data. The dependent variables were labeled X and Y, and the predictor variables were likewise labeled X and Y.

3.4 Data Normalization

A min-max scaler was used to retain value discrepancies during the translation of quantitative values into a scale-based format. Given the frequency of binary qualities, normalization was then applied to the predictor variables (X) to improve accuracy.

3.5 Data Splitting

The dataset was divided into training and testing subsets, with 80% of the data allotted for training and the remaining 20% for testing. This allocation made it easier to assess the model's predicted accuracy using the testing subset after training on the 80% dataset.

3.6 The Application of Algorithms

Twelve different techniques, including K-nearest Neighbor, Decision Tree, Random Forest, AdaBoost, XGBoost, Stochastic Gradient Descent, Linear SVC, Perceptron, Naive Bayes, Support Vector Machines, Logistic Regression, and Discriminant Analysis, were used. These approaches all resulted in various analytical revelations.

3.7 Model Assessment

The data was arranged into tables after each algorithm's performance was evaluated. Metrics such as the Confusion Matrix, Accuracy Score, Jaccard Score, Cross-Validated Score, AUC Score, Mean Absolute Error, and Mean Squared Error were used to evaluate performance. The Confusion Matrix captured the predictive potential of the data, whereas measures such as correctness Score, Jaccard Score, Cross-Validated Score, and AUC Score assessed the correctness of projected data.

3.8 Choosing the Best Algorithm

The Decision Tree Classifier was chosen as the best option after careful examination and evaluation of critical data from the tables. This method received the highest scores across all assessment criteria and was judged acceptable for the COVID-19 Clinical Dataset in this study.

4 Result Analysis

After calculating all practical components, such as Precision, Recall, F1-Measure, Accuracy, and so on, it is crucial to review the findings analysis. In this study, it will be identified which algorithm outperforms all others and which algorithm outperforms this when compared to others.

4.1 Accuracy

A crucial performance measure affected by data input is algorithm correctness. For evaluating accuracy, probabilistic approaches are frequently used. In particular, eXtreme Gradient Boosting showed the best accuracy, whereas Gaussian Naive Bayes showed the lowest. Extreme Gradient Boosting greatly improves model effectiveness and speed because to its robust and scalable nature.The best accuracy was demonstrated using eXtreme Gradient Boosting. Model performance and computing speed are enhanced by its scalability and reliability. Insights into accuracy trends and algorithm contribution are provided visually and tabularly in Figure 2 and Table 1. Figure 2 displays the accuracy curve to show model performance, and Table 1 delineates the contribution of each method to accuracy to help us better understand their influence.

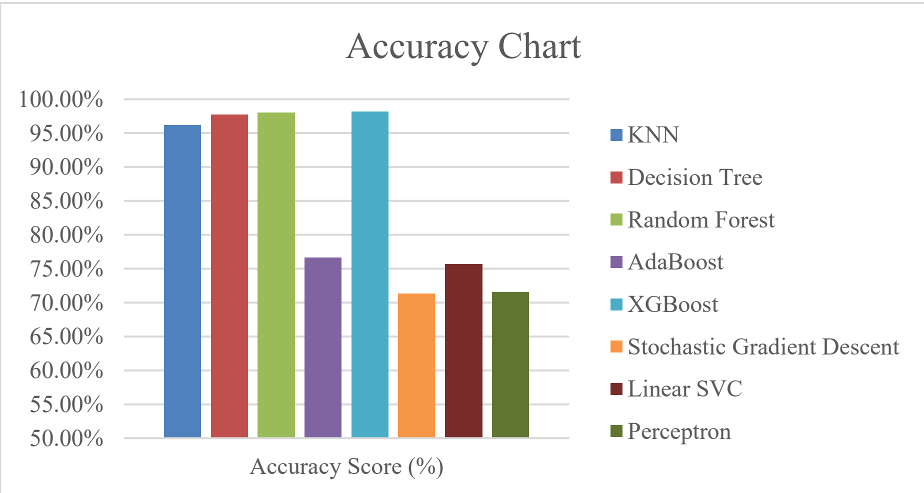

Fig. 2. Accuracy Chart

Table 1. Accuracy, Jaccard and Cross Validated Score

Algorithm Name	Accuracy Score (%)	Jaccard Score (%)	Cross Validated Score (%)
KNN	96.22%	92.71%	99.51%
Decision Tree	97.70%	95.50%	99.05%
Random Forest	98.03%	96.13%	99.67%
AdaBoost	76.64%	62.13%	73.90%
XGBoost	98.19%	96.45%	99.54%
Stochastic Gradient Descent	71.38%	55.50%	73.08%
Linear SVC	75.66%	60.85%	74.19%
Perceptron	71.55%	55.70%	67.68%

4.2 Jaccard Score

The intersection-to-union ratio is used by the Jaccard score to assess sample set similarity and diversity. It measures the overlap of two finite sets by dividing their intersection by their union, as calculated by the Jaccard coefficient. Equation (1), Fig. 3, and Table 1 give an in-depth look at accuracy trends and algorithm contributions. Equation (1) describes the Jaccard score, whereas Fig. 3 clearly depicts accuracy patterns. Table 1 depicts the distribution of forecasting methodologies, which helps us better understand their responsibilities.

$$J(A,B) = \frac{|A \cap B|}{|A \cup B|} = \frac{|A \cap B|}{|A| + |B| - |A \cap B|} \dots\dots\dots (1)$$

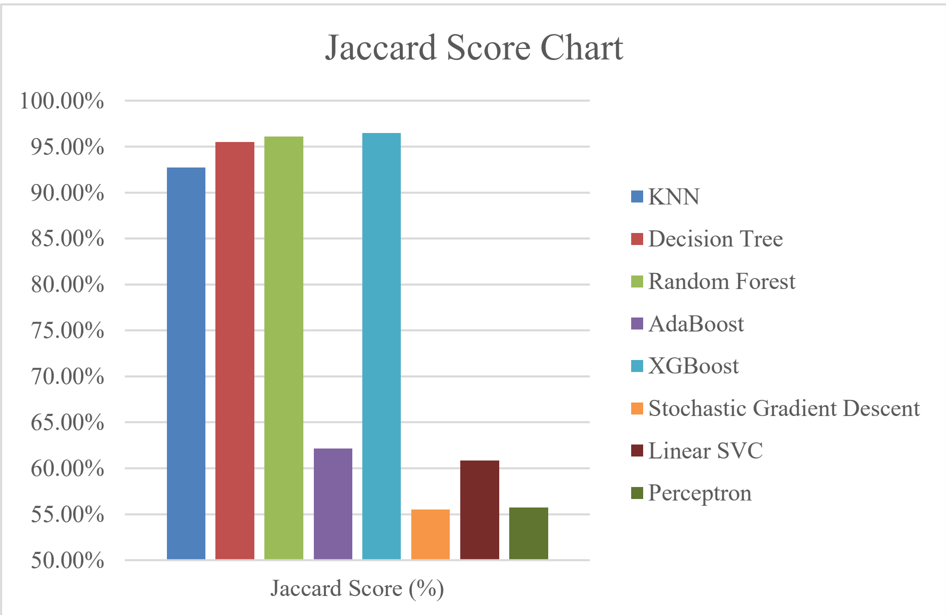

Fig. 3. Jaccard Score Chart

4.3 Cross Validated Score

A statistical approach called cross-validation is used to evaluate machine learning models. For cross-validation, the dataset is shuffled and split into k folds. A final score is calculated by averaging the (k-1)/k outcomes of each fold. For practical application, the model is then applied to the complete dataset. The cross-validated score chart and technique distribution for estimate are shown in Figure 4 and Table 1. These graphics reveal the efficiency of cross-validation and the relative contributions of several strategies in the model.

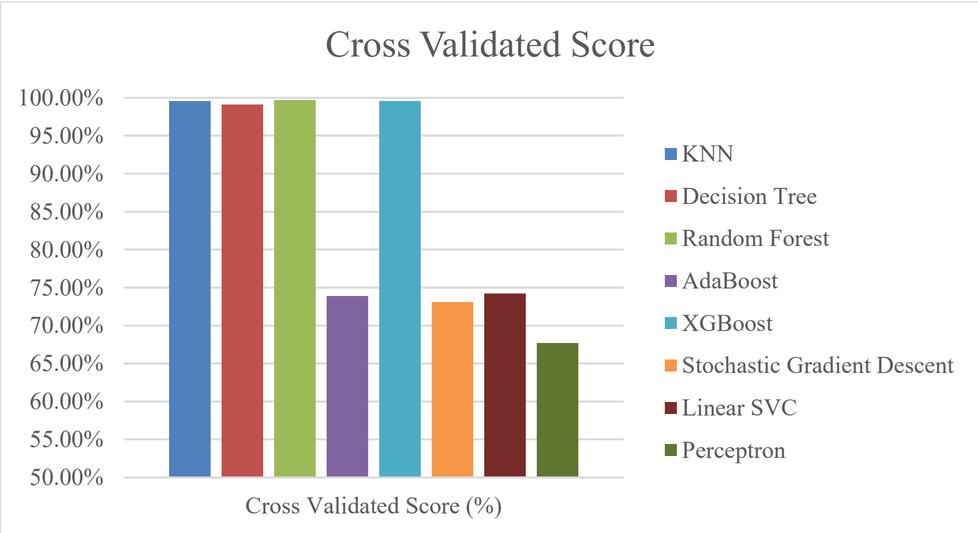

Fig. 4. Cross Validated Score Chart

5 Conclusion

To summarize, the critical importance of early discovery, proper diagnosis, and timely treatment in reducing the impact of Covid-19 on patient outcomes cannot be emphasized. However, the daunting task of producing enough Covid-19 test kits to serve a large population has resulted in diagnostic and treatment delays. The use of a model trained on a relevant dataset provides a viable route for predicting various phases of coronavirus infection.

Our study rigorously trained and tested twelve unique machine learning algorithms using crucial metrics such as accuracy, Jaccard score, Cross Validated score, and AUC scores to discover the best effective prediction model. The performance study clearly favored XGBoost over other classification choices.

A transformational approach is offered by the possible application of our paradigm throughout healthcare institutions that provide Covid-19 therapy. This approach has the potential to significantly increase the availability of Covid-19 testing by providing patients with the convenience of remotely obtaining their Covid-19 reports online, day or night, from the comfort of their homes. This comprehensive technique has the potential to not only increase testing accessibility, but also to improve early detection capabilities and, as a result, reduce the danger of viral transmission.

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